Thermal Conductivity, Heat Capacity and Thermal Expansion Measurements of Carbonaceous Chondrite CM Meteorites

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Abstract: **Measurement of the low-temperature thermodynamic and physical properties of meteorites provides fundamental data for the study and understanding of asteroids and other small bodies. Of particular interest are the CM carbonaceous chondrites, which represent a class of primitive meteorites that record substantial chemical information concerning the evolution of the early solar system. Most CM chondrites contain both anhydrous minerals such as olivine and pyroxene found in chondrules, along with abundant hydrous phyllosilicates contained in the meteorite matrix, interspersed between the chondrules. We have measured the thermal conductivity, heat capacity and thermal expansion of several CM carbonaceous chondrites (Murchison, Murray, Cold Bokkeveld, NWA 7309, Jbilet Winselwan) at low temperatures (5–300 K). The thermal expansion measurements on Murchison and Murray indicate a substantial fluctuation in meteorite volume as temperature decreases from 230 - 210 K followed by linear contraction below 210 K. Such transitions are not typical for other carbonaceous chondrites. Thermal diffusivity as a function of temperature is calculated from measurements of density, thermal conductivity and heat capacity. Our thermal diffusivity results compare well with previous estimates for similar meteorites, where conductivity was derived from diffusivity measurements and modeled heat capacities; our new values are of a higher precision and cover a wider range of temperatures.**

(1) Introduction:

(a) Heat capacity, thermal conductivity, thermal diffusivity, density, porosity and thermal expansion are the fundamental thermodynamic and physical meteoritic characteristics for modeling various aspects of the orbital and internal evolution of asteroids and small solar system bodies. These characterization techniques help to identify material and composition dynamics in the meteorite itself.

(b) While most meteoritic heat capacities are measured at T ≥ 295 K, our previous work on the shergottite Los Angeles (ref. 1) indicates that heat capacity is a strong function of temperature, especially at the lower temperatures found in the asteroid belt. Low-temperature studies of heat capacity and thermal conductivity may spark further discussion in the astronomy community and an evolution in the theory behind asteroid formation.

(c) Linear thermal expansion measurements can indicate 1st or 2nd order structural or magnetic phase transitions (Clausius-Clapyeron Eqs.) in a material. In single crystals these are usually linked with similar transitions found in specific heat capacity data (Ehrenfest Relations).

(d) These measurements were made on an automated cryogenic He^4 refrigerator, Physical Properties Measurement System (PPMS) manufactured by Quantum Design. It operates at 2 ≤ T ≤ 400 K, 0 ≤ H ≤ 90,000 Oe and calibrations are referenced to NIST standards.

(2) Specific Heat Capacity vs. T (2-300 K)

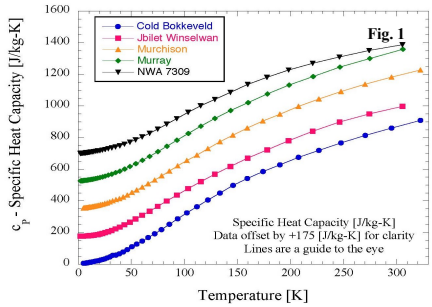

TABLE: Polynomial coefficients are calculated using the Hass-Fisher Model (ref. 2) to compare the different CM chondrite behaviors.

$$c_p = A + 2\,B\,T + C\,T^{-2} + D\,T^{2} + E\,T^{-1/2}$$

Meteorite	ρ [kg/m³]	A x 10^{-1}	B x 10^{-3}	C x 10^{0}	D x 10^{-6}	E x 10^{-1}
Cold Bokkeveld	2310	-2.77	2.88	-3.42	-7.01	8.44
Jbilet Winselwan	2040	-1.76	2.38	- 0.57	-4.98	4.09
Murchison	2320	-2.60	2.70	- 3.32	-6.21	8.03
Murray	2210	-1.63	2.30	- 0.58	-4.50	3.85
NWA 7309	2300	-1.64	2.12	- 0.58	-4.87	3.94

(3) Thermal Conductivity vs. T (5-300 K)

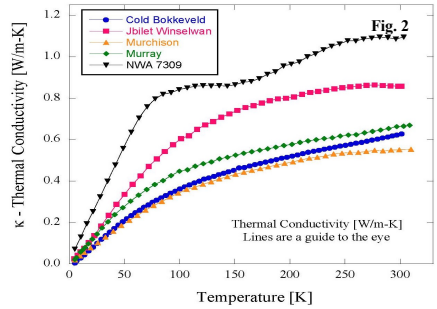

Fig. 1: Specific heat capacity was measured using a thermal relaxation technique (ref. 3) where sample dimensions were ~ 1 x 2 x 3 mm and ~ 20 mg. Our density measurements are comparable with ref. 4, 5.

Fig. 2: Thermal conductivity vs. temperature data are shown for five CM chondrite samples. The thermal conductivity sample dimensions were ~ 2 x 3 x 10 mm and measured with a cooling rate of 0.5 K/min. These data are consistent with other thermal conductivity measurements made with samples cut from different spatial orientations in the same parent meteorite (ref. 6).

(4) Thermal Diffusivity vs. T - calculation

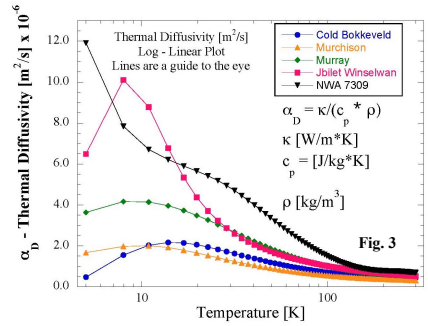

Fig. 3: Thermal diffusivity shows slight variation for T ≥ 100 K, but exhibits a significant increase at T ≤ 100 K. These data are consistent our previous thermal diffusivity measurements on ordinary (H, L) and carbonaceous (CV3) chondrites across the same temperature range. The diffusivity calculation assumes the approximation $c_V \sim c_p$

(5) Thermal Expansion vs. T (5-300 K)

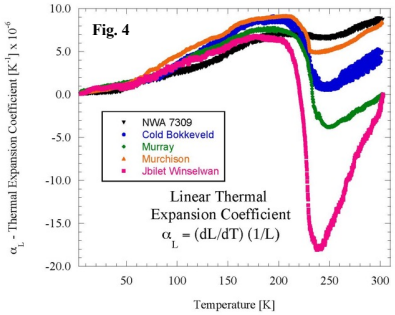

Fig. 4: Linear thermal expansion measurements indicate a second order phase transition across a narrow T range: NWA 7309 - 226.5 K, Cold Bokkeveld - 228.5 K, Murray- 231.4 K, Murchison - 227.5 K, Jbilet Winselwan - 225.6 K. Measurements were taken at a cooling rate of 0.3 K/min. on a Schmiedeshoff capacitive dilatometer (ref. 7) with a resolution of ~ 0.5 Angstrom.

(6) Conclusions/Analysis:

(a) Since the temperature dependence of thermal diffusivity has a considerable effect on thermal modeling of meteorites, significant increases in thermal diffusivity at T ≤ 100 K shown here should add to the knowledge and understanding of meteorite evolution and modeling. Further experimental data (both c_P and κ vs. T) of various mineral and meteorite species are needed to indicate if this a general phenomenon for meteorites at low T, see ref. 8.

(b) The second order phase transitions found in the narrow T range (225-232 K) in our thermal expansion data, without a concurrent transition in heat capacity, may indicate a structural transition driven by one or more of the minerals (e.g. hydrous phyllosilicates) or volatile compounds prevalent in CM carbonaceous chondrites. Future experiments on mineral species containing those compounds will shed light on the origin and nature of these 2nd order phase transitions.

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References:

(1)Opeil, *et al., Meterorit. Planet. Sci.* **47**(3) 319-329 (2012).
(2)Haas and Fisher, *Am. Jour. of Sci.* **276** April (1976).
(3) Lashley, et al., Cryogenics **43**(6) 369-378 (2003).
(4) Britt, *et al.*, *Meterorit. Planet. Sci.* **38** 1161-1180 (2003).
(5) Macke, *Ph.D. Dissertation*, U. Cent. Florida, 2010.
(6) Opeil, *et. al., Icarus* **208** 449-454 (2010).
(7) Schmiedeshoff, *et al., Rev. Sci. Inst.* **77** 123907 (2006).
(8) Ghosh and McSween, *Meteorit. Planet. Sci.* **34** 121-127 (1990).

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